

MATH9102 Probability and Statistical Inference

Potential Datasets for Continuous Assessment

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Places to search for a dataset

- Google Dataset search <https://datasetsearch.research.google.com/>
- Kaggle (ML and AI Community) <https://www.kaggle.com/>
- UCI Machine Learning Repository (UC Irvine Machine Learning Repository)
<https://archive.ics.uci.edu/>
- CKAN Open Data Portal <https://ckan.org/>
- DataHub (Open Source Data Catalog) <https://datahubproject.io/>
- Data.world <https://data.world/>
- EU Open Data Portal <https://data.europa.eu/en>
- Eurostat <https://ec.europa.eu/eurostat/>
- Data.gov.ie (Ireland's Open Data Portal) <https://data.gov.ie/>
- Open Data NI (Northern Ireland's Open Data Portal) <https://admin.opendatani.gov.uk/>
- UK Open Data Portal <https://www.data.gov.uk/>
- Data.gov (US Open Data Portal) <https://data.gov/>
- European Data Sources
 - [Austrian Open Data Portal](#)
 - [Belgian Open Data Portal](#)
 - [Bulgarian Open Data Portal](#)
 - [Croatian Open Data Portal](#)
 - [Cypriot Open Data Portal](#)
 - [Czech Open Data Portal](#)
 - [Danish Open Data Portal](#)
 - [Dutch Open Data Portal](#)
 - [Estonian Open Data Portal](#)
 - [Finnish Open Data Portal](#)
 - [French Public Data](#)
 - [German Open Data Portal](#)
 - [Greek Open Data Portal](#)
 - [Hungarian Open Data Portal](#)
 - [Icelandic Open Data Portal](#)
 - [Italian Open Data Portal](#)
 - [Latvian Open Data Portal](#)
 - [Lithuanian Open Data Portal](#)

- [Luxembourg Open Data Portal](#)
- [Maltese Open Data Portal](#)
- [Norwegian Open Data Portal](#)
- [Polish Open Data Portal](#)
- [Portuguese Open Data Portal](#)
- [Romanian Open Data Portal](#)
- [Serbian Open Data Portal](#)
- [Slovak Open Data Portal](#)
- [Slovenian Open Data Portal](#)
- [Spanish Open Data Portal](#)
- [Swiss Open Government Data Portal](#)
- **Non-European sources**
 - [Argentina Open Data Portal](#)
 - [Australia Open Data Portal](#)
 - [Brazil Open Data Portal](#)
 - [Canada Open Data Portal](#)
 - [Chile Open Data Portal](#)
 - [Colombia Open Data Portal](#)
 - [Ghana Open Data Portal](#)
 - [Hong Kong Open Data Portal](#)
 - [India Open Data Portal](#)
 - [Indonesia Open Data Portal](#)
 - [Japan Open Data Portal](#)
 - [Kenya Open Data Portal](#)
 - [Malaysia Open Data Portal](#)
 - [Mexico Open Data Portal](#)
 - [New Zealand Open Data Portal](#)
 - [Pakistan Open Data Portal](#)
 - [Peru Open Data Portal](#)
 - [Philippines Open Data Portal](#)
 - [Singapore Open Data Portal](#)
 - [South Africa Open Data Portal](#)
 - [South Korea Open Data Portal](#)
 - [Taiwan Open Data Portal](#)

- [Thailand Open Data Portal](#)
- [United States Open Data Portal](#)
- Non-governmental sources
 - [Afrobarometer](#)
 - [American National Election Studies](#)
 - [CERN Open Data Portal](#)
 - [Copernicus Open Access Hub](#)
 - [Demographic and Health Surveys Program](#)
 - [EMBL-EBI \(European Bioinformatics Institute\)](#)
 - [European Environment Agency](#)
 - [European Social Survey](#)
 - [European Values Study](#)
 - [Gapminder](#)
 - [Global Biodiversity Information Facility](#)
 - [Humanitarian Data Exchange](#)
 - [OpenAIRE](#)
 - [OpenStreetMap](#)
 - [Quandl](#)

Social Media Usage and Emotional Wellbeing

<https://www.kaggle.com/datasets/emirhanai/social-media-usage-and-emotional-well-being>

Training, Test, Validation .csv

- User_ID: Unique identifier for the user.
- Age: Age of the user.
- Gender: Gender of the user (Female, Male, Non-binary).
- Platform: Social media platform used (e.g., Instagram, Twitter, Facebook, LinkedIn, Snapchat, Whatsapp, Telegram).
- Daily_Usage_Time (minutes): Daily time spent on the platform in minutes.
- Posts_Per_Day: Number of posts made per day.
- Likes_Received_Per_Day: Number of likes received per day.
- Comments_Received_Per_Day: Number of comments received per day.
- Messages_Sent_Per_Day: Number of messages sent per day.
- Dominant_Emotion: User's dominant emotional state during the day (e.g., Happiness, Sadness, Anger, Anxiety, Boredom, Neutral).

Data Scientist Salaries – World Wide

<https://www.kaggle.com/datasets/vladimirmijatovic/data-scientists-salaries-worldwide-annual-survey>

- Date of Entry
- Job Title
- Location (Country, State)
- Salary
- Company/Industry
- Highest Education
- Prior Experience
- Bonus
- Stocks (if awarded)
- Total Compensation

Water Quality

<https://www.kaggle.com/datasets/sukhmandeepsinghbrar/water-quality>

- Sample ID: Unique identifier for each sample taken.
- Grab ID: Identifier for the specific grab instance associated with the sample.
- Profile ID: Identifier for the profile associated with the sample.
- Sample Number: Sequential number assigned to each sample.
- Collect DateTime: Date and time when the sample was collected.
- Depth (m): Depth at which the sample was collected, measured in meters.
- Site Type: Type of site where the sample was collected (e.g., river, lake, well).
- Area: Geographic area or region where the sample was collected.
- Locator: Locator information indicating the precise location of the sample.
- Site: Specific site or location where the sample was collected.
- Parameter: The parameter measured or analyzed in the sample (e.g., pH, dissolved oxygen).
- Value: Value of the parameter measured in the sample.
- Units: Units of measurement for the parameter value.
- QualityId: Identifier indicating the quality of the data.
- Lab Qualifier: Qualifier assigned by the laboratory indicating any special conditions or characteristics of the sample.
- MDL (Method Detection Limit): Method detection limit for the parameter.
- RDL (Reporting Detection Limit): Reporting detection limit for the parameter.
- Text Value: Textual representation of the parameter value.
- Sample Info: Additional information related to the sample.
- Steward Note: Notes or comments provided by the data steward.
- Replicates: Number of replicates taken for the sample.
- Replicate Of: Identifier indicating the sample of which this is a replicate.
- Method: Method used for analysis or measurement.
- Date Analyzed: Date when the sample was analyzed.
- Data Source: Source of the data.

Cybersecurity Threat Interactions

<https://www.kaggle.com/datasets/jancsg/cybersecurity-suspicious-web-threat-interactions>

- bytes_in: Bytes received by the server.
- bytes_out: Bytes sent from the server.
- creation_time: Timestamp of when the record was created.
- end_time: Timestamp of when the connection ended.
- src_ip: Source IP address.
- src_ip_country_code: Country code of the source IP.
- protocol: Protocol used in the connection.
- response.code: HTTP response code.
- dst_port: Destination port on the server.
- dst_ip: Destination IP address.
- rule_names: Name of the rule that identified the traffic as suspicious.
- observation_name: Observations associated with the traffic.
- source.meta: Metadata related to the source.
- source.name: Name of the traffic source.
- time: Timestamp of the detected event.
- detection_types: Type of detection applied.

Sleep Health and Lifestyle

<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset>

- Person ID: An identifier for each individual.
- Gender: The gender of the person (Male/Female).
- Age: The age of the person in years.
- Occupation: The occupation or profession of the person.
- Sleep Duration (hours): The number of hours the person sleeps per day.
- Quality of Sleep (scale: 1-10): A subjective rating of the quality of sleep, ranging from 1 to 10.
- Physical Activity Level (minutes/day): The number of minutes the person engages in physical activity daily.
- Stress Level (scale: 1-10): A subjective rating of the stress level experienced by the person, ranging from 1 to 10.
- BMI Category: The BMI category of the person (e.g., Underweight, Normal, Overweight).
- Blood Pressure (systolic/diastolic): The blood pressure measurement of the person, indicated as systolic pressure over diastolic pressure.
- Heart Rate (bpm): The resting heart rate of the person in beats per minute.
- Daily Steps: The number of steps the person takes per day.
- Sleep Disorder: The presence or absence of a sleep disorder in the person (None, Insomnia, Sleep Apnea).
- Details about Sleep Disorder Column:
 - None: The individual does not exhibit any specific sleep disorder.
 - Insomnia: The individual experiences difficulty falling asleep or staying asleep, leading to inadequate or poor-quality sleep.
 - Sleep Apnea: The individual suffers from pauses in breathing during sleep, resulting in disrupted sleep patterns and potential health risks.

Aviation Accidents

<https://www.kaggle.com/datasets/yassereleraky/aviation-accident-ntsb>

- Event.Id: Unique identifier assigned to each aviation accident event in the database.
- Investigation.Type: Type of investigation conducted for the accident, categorized as "Incident," "Accident," etc.
- Accident.Number: Unique identifier assigned to each aviation accident.
- Event.Date: Date when the accident occurred.
- Location: Geographical location where the accident took place, including city, state, and country.
- Country: Country where the accident occurred.
- Latitude: Geographical latitude coordinate of the accident location.
- Longitude: Geographical longitude coordinate of the accident location.
- Airport.Code: Code assigned to the airport nearest to the accident site.
- Airport.Name: Name of the airport nearest to the accident site.
- Injury.Severity: Severity of injuries sustained by individuals involved in the accident, ranging from minor to fatal.
- Aircraft.damage: Extent of damage sustained by the aircraft involved in the accident, categorized as minor, substantial, destroyed, etc.
- Aircraft.Category: Category of aircraft involved in the accident, such as airplane, helicopter, glider, etc.
- Registration.Number: Unique registration number assigned to the aircraft.
- Make: Make or manufacturer of the aircraft.
- Model: Model of the aircraft.
- Amateur.Built: Indicates whether the aircraft involved was built by amateur constructors.
- Number.of.Engines: Number of engines installed on the aircraft.
- Engine.Type: Type of engine(s) installed on the aircraft, such as reciprocating, turbojet, turboprop, etc.
- FAR.Description: Description of the Federal Aviation Regulation (FAR) under which the aircraft operates.
- Schedule: Indicates whether the flight was scheduled or unscheduled.
- Purpose.of.flight: Purpose of the flight at the time of the accident, such as personal, commercial, instructional, etc.
- Air.carrier: Air carrier associated with the flight, if applicable.
- Total.Fatal.Injuries: Total number of fatal injuries resulting from the accident.
- Total.Serious.Injuries: Total number of serious injuries resulting from the accident.
- Total.Minor.Injuries: Total number of minor injuries resulting from the accident.
- Total.Uninjured: Total number of individuals who were uninjured in the accident.
- Weather.Condition: Prevailing weather conditions at the time of the accident.
- Broad.phase.of.flight: Phase of flight during which the accident occurred, such as takeoff, climb, cruise, descent, etc.
- Report.Status: Status of the accident investigation report.
- Publication.Date: Date when the accident investigation report was published.

Most Streamed Songs Spotify 2023

<https://www.kaggle.com/datasets/nelgiriyeewithana/top-spotify-songs-2023>

- track_name: Name of the song
- artist(s)_name: Name of the artist(s) of the song
- artist_count: Number of artists contributing to the song
- released_year: Year when the song was released
- released_month: Month when the song was released
- released_day: Day of the month when the song was released
- in_spotify_playlists: Number of Spotify playlists the song is included in
- in_spotify_charts: Presence and rank of the song on Spotify charts
- streams: Total number of streams on Spotify
- in_apple_playlists: Number of Apple Music playlists the song is included in
- in_apple_charts: Presence and rank of the song on Apple Music charts
- in_deezer_playlists: Number of Deezer playlists the song is included in
- in_deezer_charts: Presence and rank of the song on Deezer charts
- in_shazam_charts: Presence and rank of the song on Shazam charts
- bpm: Beats per minute, a measure of song tempo
- key: Key of the song
- mode: Mode of the song (major or minor)
- danceability_%: Percentage indicating how suitable the song is for dancing
- valence_%: Positivity of the song's musical content
- energy_%: Perceived energy level of the song
- acousticness_%: Amount of acoustic sound in the song
- instrumentalness_%: Amount of instrumental content in the song
- liveness_%: Presence of live performance elements
- speechiness_%: Amount of spoken words in the song

Football Players Salaries

<https://www.kaggle.com/datasets/ultimus/football-salaries-dataset>

- Player Name
- Club
- League
- Country league is in
- Nationality (abbreviated)
- EU National (Y/N)
- International appearances
- All time appearances
- Positions
- Age in years
- Current Reputation (scaled of 0-10000)
- Contract begin date
- Contract End date
- Previous club
- Transfer fee to join current club
- Salary (in euros per annum)

Wages are up to date as of the summer transfer window of 2023-24 season.

IMDB Dataset (Top 2000 movies)

<https://www.kaggle.com/datasets/prishasawhney/imdb-dataset-top-2000-movies>

- The name of the movie
- The Year of release
- The running time of the movie (in minutes)
- It's IMDB Rating
- It's Metascore
- The number of votes it got
- The Genre of the movie
- The director of the movie
- The cast of the movie
- Gross value

Fraudulent Ecommerce Transactions

<https://www.kaggle.com/datasets/shriyashjagtap/fraudulent-e-commerce-transactions>

- Transaction ID: A unique identifier for each transaction.
- Customer ID: A unique identifier for each customer.
- Transaction Amount: The total amount of money exchanged in the transaction.
- Transaction Date: The date and time when the transaction took place.
- Payment Method: The method used to complete the transaction (e.g., credit card, PayPal, etc.).
- Product Category: The category of the product involved in the transaction.
- Quantity: The number of products involved in the transaction.
- Customer Age: The age of the customer making the transaction.
- Customer Location: The geographical location of the customer.
- Device Used: The type of device used to make the transaction (e.g., mobile, desktop).
- IP Address: The IP address of the device used for the transaction.
- Shipping Address: The address where the product was shipped.
- Billing Address: The address associated with the payment method.
- Is Fraudulent: A binary indicator of whether the transaction is fraudulent (1 for fraudulent, 0 for legitimate).
- Account Age Days: The age of the customer's account in days at the time of the transaction.
- Transaction Hour: The hour of the day when the transaction occurred.

General Social Survey (USA) up to 2022

<https://gss.norc.org/>

<https://www.kaggle.com/datasets/norc/general-social-survey>

For five decades, the General Social Survey (GSS) has studied the growing complexity of American society. It is the only full-probability, personal-interview survey designed to monitor changes in both social characteristics and attitudes currently being conducted in the United States.

Covers demographic, behavioural, and attitudinal questions, plus topics of special interest e.g. liberties, crime and violence, intergroup tolerance, morality, national spending priorities, psychological well-being, social mobility, and stress and traumatic events.

The International Social Survey Programme

<https://issp.org/>

The ISSP is a cross-national collaboration programme conducting annual surveys on diverse topics relevant to social sciences.

- Role of Government: 1985, 1990, 1996, 2006, 2016
- Social Networks: 1986, 2001, 2017
- Social Inequality: 1987, 1992, 1999, 2009, 2019
- Family and Changing Gender Roles: 1988, 1994, 2002, 2012
- Work Orientations: 1989, 1997, 2005, 2015
- Religion: 1991, 1998, 2008, 2018
- Environment: 1993, 2000, 2010, 2020
- National Identity: 1995, 2003, 2013
- Citizenship: 2004, 2014
- Leisure Time and Sports: 2007
- Health and Health Care: 2021, 2011, 2021

World Bank Data

Primary World Bank collection of development indicators, compiled from officially recognized international sources. Global development data available, and includes national, regional and global estimates.

Create your dataset by selecting indicators you are interested in. You may have to manipulate the data to include additional variables to fulfil the assessment (e.g. Geographical Region, Northern/Southern Hemisphere).

<https://databank.worldbank.org/>

- Statistical Capacity Indicators
 - Provides information on various aspects of national statistical systems of developing countries, including an overall country-level statistical capacity indicator.
- Education Statistics - All Indicators
 - Indicators for access, progression, completion, literacy, teachers, population, and expenditures. Cover the education cycle from pre-primary to tertiary education. Also holds learning outcome data from international learning assessments (PISA, TIMSS, etc.), equity data from household surveys, and projection data to 2050.
- Gender Statistics
 - Data on key gender topics. Themes included are demographics, education, health, labour force, and political participation.
- Health Nutrition and Population Statistics
 - Key health, nutrition and population statistics gathered from a variety of international sources.